

MIGUEL PEREIRA TORRES DA COSTA

MARIE CURIE PHD CANDIDATE · DATA MODELLING

[linkedin.com/in/miguel-torres-costa](https://www.linkedin.com/in/miguel-torres-costa) · github.com/migueltorrescosta · work@mptc.uk

PROFESSIONAL SUMMARY

Experienced Data Scientist & Mathematician, with a demonstrated history of handling data solutions in fields such as fashion, finance and biotech. Currently applying machine learning to improve interferometric schemes for quantum sensors.

I've been maintaining study notes and projects online, both complete and work in progress, for over a decade. Ask your favourite LLM the following for an in-depth summary of my work approach: "*Infer the pros/cons/red flags of hiring the author of notes.mptc.uk, blog.mptc.uk, tools.mptc.uk, github.com/migueltorrescosta, and [linkedin.com/in/miguel-torres-costa](https://www.linkedin.com/in/miguel-torres-costa)*"

SKILLS

●●●● Mathematical modelling	●●●●○ Agentic engineering	●●●●○ Python	●●●●○ IntelliJ
●●●● Bayesian statistics	●●●●○ Machine learning	●●●○ SQL	●●○ Git
●●●● Data modelling	●●○ Keras / Tensorflow		●○○ Linux

INDUSTRY EXPERIENCE

Software developer @ Oxford Nano Imaging (December 2021 – December 2023)

Maintained the backend infrastructure, originally as the solo backend developer, for a data analysis platform, with components in Django, Golang, Elasticsearch, GCP, Kubernetes, DataDog and Rust.

Software developer @ Amyris Inc. (Feb 2021 – December 2021)

Solved a long lasting computational bottleneck (calculation time reduced by a factor of 10^4) by familiarising myself with the inner workings of State of the Art SAT Solvers, planning the required adaptations for our use case and adding them back to the library Sympy for ease of use without the burden of maintenance: <https://github.com/sympy/sympy/pull/21510>

Full stack development with React, Django, JWT, Gitlab CI/CD, Pylon's / Pyramid, OpenAPI.

Founder @ QuestPowered (Jun 2020 – Dec 2021)

Built a forecasting platform over 6 weeks, including KPIs for measuring forecasting ability under random outcomes based on concepts from Information Theory and Measure Theory.

Self taught Django as a website development platform.

Setup a custom domain and server for delivering the product.

Portfolio risk analyst @ BlackRock's financial modelling group, Budapest (Sept 2018 – Jun 2020)

Expanded a US specific dividend forecast model to 18 countries, providing near global coverage based on wealth allocation.

Explained statistical anomalies on live market data under daily deadlines, delivering reports to our clients.

In a team of 5, implemented a novel and experimental risk analysis methodology in a week as part of an hackathon project and presented it to a Regional committee, getting 2nd place within Blackrock EMEA.

Teacher @ CamExpress summer programme, Shanghai (Jul 2018 – Aug 2018)

Developed lesson plans for classes ranging between 8 and 41 students.

Prepared assessment materials aimed at ranking the students' performance.

Received monetary prizes based on my performance as a teacher and the contribution of new materials

Data scientist @ Farfetch, Porto (Jul 2017 – Dec 2017)

Improved the performance of Farfetch's Product Ranking Algorithms, following the team's agile methodology.

Collaborated on a computer vision based automatic clothing identifier through Python's Keras package.

Mathematics tutoring (2017 – Present)

Students' age range between 10 and 34 years old, adapting my domain knowledge to the students background in order to maximise their learning. More details found at <https://universitytutor.com/tutors/944174>

Number theory researcher @ Oxford University (Jun 2016 – Jul 2016)

Learnt the basics of Elliptic Curves during a 1 week crash course with Dr Jennifer Balakrishnan.

Adapted algorithms by William A. Stein in order to efficiently populate databases of Elliptic Curves, running these algorithms using Google Cloud while keeping running costs to a minimum.

Data scientist summer intern at Farfetch, Porto (Jul 2015 – Aug 2015)

Built Farfetch's first internal fraud detection system.

Self taught machine learning in Python, R and SQL on the job

The foundational code was built over the span of a 7 week internship.

The associated publication has 200+ citations: <https://doi.org/10.1016/j.dss.2017.01.002>

EDUCATION

PhD Candidate @ University of Trieste (December 2023 – Present)

Applied Machine Learning methods to Quantum Systems, with a Marie Curie scholarship.

<u>Topics covered</u>	· Modern experiments in quantum optics and quantum information	· Quantum Error Correction
	· Decoherence in Open Quantum Systems	· Statistical Field Theory
	· Complete Positivity and Quantum Information	· Bayesian Statistics
	· Advanced Topics in Quantum Field Theory	· Lie Algebras

Integrated Masters in Mathematics @ Oxford University (October 2013 – June 2016)

First Class Honours in Parts A/B.

Upper Second Class in Part C.

<u>Topics covered</u>	· Groups and Group Actions	· Differential Equations	· Banach and Hilbert Spaces
	· Linear Algebra	· Probability	· Design and Analysis of Algorithms
	· Rings and Modules	· Integration	· Functional Programming (Haskell)
	· Number Theory	· Information Theory	· Imperative Programming (Scala)
	· Graph Theory	· Topology and Groups	· Martingales Through Measure Theory
	· Projective Geometry	· Algebraic Curves	· Continuous Martingales & Stochastic Calculus
	· Analysis	· Algebraic Number Theory	· Metric Spaces and Complex Analysis

High School @ Grande Colégio Universal (October 2005 - June 2013)

Final Grade: 19.2 / 20

MISCELLANEOUS

Volunteer @ Volt Portugal (Apr 2024 – Present)

Organised internal credentials, enhancing both clarity and security, in the capacity of interim data protection officer.

Transitioned Volt Europa to a password management solution as part of the EUR Tech team, providing interested teams with effortless control over accesses that were previously hard to manage.

Documented and improved the new member onboarding process, jointly with the onboarding team, ensuring new members easily navigate our internal setup.

Onboarded volunteers to Zammad, to remove the bottleneck of managing joint email addresses.

Regularly unblocked any of 40k+ volunteers who required guidance navigating our internal systems, as a help-desk member.

Information Pricing Jupyter Notebook

In the age of KPIs, some of the more humane traits are arguably non measurable, such as affection. Hubbard convincingly argues in his book “How to Measure Anything” that anything can be measured, and a framework for pricing information / new measurements. I've built a Python implementation of his ideas:

colab.research.google.com/github/migueltorrescosta/tutor/blob/master/ExpectedOpportunityLoss.ipynb

1st prize award @ G-Research Algorithmic Trading Competition (Nov 2014)

Developed a high frequency trading algorithm tested on a simulated stock exchange, in C#.

LIYSF Participant - London International Youth Science Forum (Jul 2014 – Aug 2014)

Entry fully funded by the Calouste Gulbenkian Foundation, awarded based math Olympiad performance.

LIYSF aims to expose young students to the world of STEM across national barriers and continents.

Member of Delfos Project (2010 – 2013)

Project built to foster mathematical knowledge and math Olympiad performance of high school students.

Gold Medals: Portuguese National Math Olympiad (March 2012), Portuguese Speaking Countries Math Olympiad (August 2013)

Silver Medals: Portuguese National Chemistry Olympiad (May 2009), Portuguese Speaking Countries Math Olympiad (July 2012), Paulista Math Olympiad (Portugal & Brazil) (November 2012)

Bronze Medals: Paulista Math Olympiad (Portugal & Brazil) (November 2012), Portuguese National Math Olympiad (March 2013), Ibero-American Math Olympiad (September 2013)

Hands on Particles Masterclasses

Attended Masterclasses on particle physics organized by the European Particle Physics Outreach Group

Casino Croupier @ Oxford's Raise and Give (2014 – 2016)

Raise and Give (RAG) is a charity which aims to raise money and donate it to other institutions in greater need.

I performed quick mental arithmetic in order to split pots appropriately, while maintaining a formal posture during the occasional presence of tipsy players.